

Math 6601 Homework 7

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Problem 1.

A linear model has the form $y = ax + b$, with parameters a and b . If data $\{(x_i, y_i)\}_{i=1}^n$ are given, then to fit this model with least squares, we want to find the least-squares solution of $Ac = y$, where

$$A = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad c = \begin{bmatrix} b \\ a \end{bmatrix}.$$

Then $Ac = y$ is equivalent to $ax_i + b = y_i$ for $i = 1, 2, \dots, n$.

(a) Based on the data, we have

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix}, \quad y = \begin{bmatrix} 0 \\ 3 \\ 3 \\ 6 \end{bmatrix}.$$

Then the least squares solution of $Ac = y$ is the solution c' of $(A^T A)c = A^T y$, the normal equations. We have

$$A^T A = \begin{bmatrix} 4 & 8 \\ 8 & 30 \end{bmatrix}, \quad A^T y = \begin{bmatrix} 12 \\ 39 \end{bmatrix}.$$

Then

$$(A^T A)^{-1} = \frac{1}{56} \begin{bmatrix} 30 & -8 \\ -8 & 4 \end{bmatrix},$$

and

$$c' = (A^T A)^{-1} A^T y = \frac{1}{56} \begin{bmatrix} 30 & -8 \\ -8 & 4 \end{bmatrix} \begin{bmatrix} 12 \\ 39 \end{bmatrix} = \frac{1}{56} \begin{bmatrix} 48 \\ 60 \end{bmatrix}.$$

Thus, $b = \frac{6}{7}$ and $a = \frac{15}{14}$ are the parameters needed.

(b) Based on the data, we have

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad y = \begin{bmatrix} 5 \\ 3 \\ 3 \\ 1 \end{bmatrix}.$$

Then the least squares solution of $Ac = y$ is the solution c' of $(A^T A)c = A^T y$, the normal equations. We have

$$A^T A = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}, \quad A^T y = \begin{bmatrix} 12 \\ 12 \end{bmatrix}.$$

Then

$$(A^T A)^{-1} = \frac{1}{20} \begin{bmatrix} 14 & -6 \\ -6 & 4 \end{bmatrix},$$

and

$$c' = (A^T A)^{-1} A^T y = \frac{1}{10} \begin{bmatrix} 7 & -3 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 12 \\ 12 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 48 \\ -12 \end{bmatrix}.$$

Thus, $b = \frac{24}{5}$ and $a = -\frac{6}{5}$ are the parameters needed.

Problem 2.

A parabolic model has the form $y = ax^2 + bx + c$, with parameters a , b , and c . If data $\{(x_i, y_i)\}_{i=1}^n$ are given, then to fit this model with least squares, we want to find the least-squares solution of $Ad = y$, where

$$A = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad d = \begin{bmatrix} c \\ b \\ a \end{bmatrix}.$$

Then $Ad = y$ is equivalent to $ax_i^2 + bx_i + c = y_i$ for $i = 1, 2, \dots, n$.

(a) From the data we get

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 5 & 25 \end{bmatrix}, \quad y = \begin{bmatrix} 0 \\ 3 \\ 3 \\ 6 \end{bmatrix}.$$

We need to solve $(A^T A)d = A^T y$. We have

$$A^T A = \begin{bmatrix} 4 & 8 & 30 \\ 8 & 30 & 134 \\ 30 & 134 & 642 \end{bmatrix}, \quad A^T y = \begin{bmatrix} 12 \\ 39 \\ 165 \end{bmatrix}.$$

Solving for $d = (A^T A)^{-1} A^T y$ gives

$$d = \frac{1}{362} \begin{bmatrix} 126 \\ 705 \\ -60 \end{bmatrix},$$

so $c = \frac{63}{181}$, $b = \frac{705}{362}$, and $a = -\frac{30}{181}$ are the desired parameters.

(b) From the data we get

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}, \quad y = \begin{bmatrix} 5 \\ 3 \\ 3 \\ 1 \end{bmatrix}.$$

We need to solve $(A^T A)d = A^T y$. We have

$$A^T A = \begin{bmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{bmatrix}, \quad A^T y = \begin{bmatrix} 12 \\ 12 \\ 24 \end{bmatrix}.$$

Solving for $d = (A^T A)^{-1} A^T y$ gives

$$d = \begin{bmatrix} \frac{24}{5} \\ -\frac{6}{5} \\ 0 \end{bmatrix},$$

so $c = \frac{24}{5}$, $b = -\frac{6}{5}$, and $a = 0$ are the desired parameters.

Problem 3.

A third-degree polynomial model has the form $y = ax^3 + bx^2 + cx + d$, with parameters a , b , c , and d . If data $\{(x_i, y_i)\}_{i=1}^n$ are given, then to fit this model with least squares, we want to find the least-squares solution of $Ae = y$, where

$$A = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 & x_n^3 \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad e = \begin{bmatrix} d \\ c \\ b \\ a \end{bmatrix}.$$

Then $Ae = y$ is equivalent to $ax_i^3 + bx_i^2 + cx_i + d = y_i$ for $i = 1, 2, \dots, n$. From the data we get

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 5 & 25 & 125 \end{bmatrix}, \quad y = \begin{bmatrix} 0 \\ 3 \\ 3 \\ 6 \end{bmatrix}.$$

We need to solve $(A^T A)d = A^T y$. We have

$$A^T A = \begin{bmatrix} 4 & 8 & 30 & 134 \\ 8 & 30 & 134 & 642 \\ 30 & 134 & 642 & 3158 \\ 134 & 642 & 3158 & 15690 \end{bmatrix}, \quad A^T y = \begin{bmatrix} 12 \\ 39 \\ 165 \\ 777 \end{bmatrix}.$$

Solving for $e = (A^T A)^{-1} A^T y$ gives

$$e = \frac{1}{20} \begin{bmatrix} 0 \\ 52 \\ -51 \\ 7 \end{bmatrix},$$

so $a = \frac{7}{20}$, $b = -\frac{51}{20}$, $c = \frac{26}{5}$, and $d = 0$ are the desired parameters.

To find the degree-3 interpolating polynomial for the data, we just need to solve $Ae = y$; since A is invertible (it is a Vandermonde matrix), this is equivalent to solving $(A^T A)e = A^T y$, so we get the exact same polynomial for both the least-squares and interpolating polynomials.

Problem 4.

If data $\{(t_i, t_i)\}_{i=1}^n$ are given, then we want to find the least-squares solution of $Ac = y$, where

$$A = \begin{bmatrix} 1 & \cos 2\pi t_1 & \sin 2\pi t_1 \\ 1 & \cos 2\pi t_2 & \sin 2\pi t_2 \\ \vdots & \vdots & \vdots \\ 1 & \cos 2\pi t_n & \sin 2\pi t_n \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad c = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Then $Ac = y$ is equivalent to $c_1 + c_2 \cos 2\pi t_i + c_3 \sin 2\pi t_i = y_i$ for $i = 1, 2, \dots, n$. From the given data we get

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 1 & 0 & 1 \\ 1 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 0 \end{bmatrix}.$$

We need to solve $(A^T A)c = A^T y$ to find the least-squares solution c . We have

$$A^T A = \begin{bmatrix} 4 & 1 & 1 + \sqrt{2} \\ 1 & 2 & 0 \\ 1 + \sqrt{2} & 0 & 2 \end{bmatrix}, \quad A^T y = \begin{bmatrix} 6 \\ 1 + \frac{3\sqrt{2}}{2} \\ 2 + \frac{3\sqrt{2}}{2} \end{bmatrix}.$$

Then $2c_2 = -c_1 + 1 + \frac{3\sqrt{2}}{2}$ by the middle row of $(A^T A)c = A^T y$, and $2c_3 = -(1 + \sqrt{2})c_1 + 2 + \frac{3\sqrt{2}}{2}$ by the third row. Substituting into the first row gives

$$8c_1 - c_1 + 1 + \frac{3\sqrt{2}}{2} - (1 + \sqrt{2})^2 c_1 + (1 + \sqrt{2}) \left(2 + \frac{3\sqrt{2}}{2} \right) = 12.$$

Then

$$(4 - 2\sqrt{2})c_1 = 12 - 1 - \frac{3\sqrt{2}}{2} - 2 - 2\sqrt{2} - \frac{3\sqrt{2}}{2} - 3 = 6 - 5\sqrt{2}.$$

Thus,

$$c_1 = \frac{6 - 5\sqrt{2}}{4 - 2\sqrt{2}} = \frac{(6 - 5\sqrt{2})(2 + \sqrt{2})}{2(4 - 2)} = \frac{1}{2} - \sqrt{2},$$

and

$$c_2 = -\frac{1}{2}c_1 + \frac{1}{2} + \frac{3\sqrt{2}}{4} = \frac{1}{4} + \frac{5\sqrt{2}}{4},$$

and

$$c_3 = -\frac{1 + \sqrt{2}}{2}c_1 + 1 + \frac{3\sqrt{2}}{4} = \frac{7}{4} + \sqrt{2}$$